# Operating Instructions

Gas engine-generator sets

**MTU 8V4000 GS – GG08V4000A1/A2 with engine 8V4000L64**
**MTU 8V4000 GS – GG08V4000A1/A2 with engine 8V4000L64FNER**
**MTU 12V4000 GS – GG12V4000A1/A2 with engine 12V4000L64**
**MTU 12V4000 GS – GG12V4000A1/A2 with engine 12V4000L64FNER**
**MTU 16V4000 GS – GG16V4000A1/A2 with engine 16V4000L64**
**MTU 16V4000 GS – GG16V4000A1/A2 with engine 16V4000L64FNER**
**MTU 20V4000 GS – GG20V4000A1/A2 with engine 20V4000L64**
**MTU 20V4000 GS – GG20V4000A1/A2 with engine 20V4000L64FNER**

SS180042/04E

A Rolls-Royce solution

| Name | Model number | Engine model | Application group |
|---|---|---|---|
| MTU 8V4000 GS | GG08V4000A1 | 8V4000L64<br>8V4000L64FNER | 3A, continuous operation, unrestricted |
| MTU 8V4000 GS | GG08V4000A2 | 8V4000L64<br>8V4000L64FNER | 3A, continuous operation, unrestricted |
| MTU 12V4000 GS | GG12V4000A1 | 12V4000L64<br>12V4000L64FNER | 3A, continuous operation, unrestricted |
| MTU 12V4000 GS | GG12V4000A2 | 12V4000L64<br>12V4000L64FNER | 3A, continuous operation, unrestricted |
| MTU 16V4000 GS | GG16V4000A1 | 16V4000L64<br>16V4000L64FNER | 3A, continuous operation, unrestricted |
| MTU 16V4000 GS | GG16V4000A2 | 16V4000L64<br>16V4000L64FNER | 3A, continuous operation, unrestricted |
| MTU 20V4000 GS | GG20V4000A1 | 20V4000L64<br>20V4000L64FNER | 3A, continuous operation, unrestricted |
| MTU 20V4000 GS | GG20V4000A2 | 20V4000L64<br>20V4000L64FNER | 3A, continuous operation, unrestricted |

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# 1 Preface

## 1.1 Preface

These Operating Instructions contain general instructions for the proper operation of your engine-generator set from the manufacturer MTU.It is vitally important that all persons working on or with the engine-generator set are familiar with all the contents of this manual and follow the instructions provided carefully at all times.

In some cases, it may prove necessary to depart from the guidelines specified in this manual in response to extraneous circumstances. Any such departure must comply with locally applicable directives and standards. All relevant safety directives and safety precautions shall be duly considered and respected.

Following these Operating Instructions will ensure that the plant operates efficiently and reliably.

All procedural instructions and safety precautions shall be followed closely to ensure that the plant operates as intended and to avoid injury or harm. Read and follow the instructions in the “Safety” section at the beginning of this manual.

All the instructions and diagrams have been checked to ensure accuracy and user-friendliness, however the skills and qualification of the operating personnel remain the most significant factor nevertheless. MTU cannot give any assurances for the actual results of any work performed according to the descriptions provided in this manual, nor shall MTU be held liable for any resultant personal injury or material damage. Personnel charged with running and operating the plant do so entirely at their own risk.

Other documents included in the scope of supply shall also be given due consideration in addition to these Operating Instructions.

All documents required by Machinery Directive 2006/42/EC are included in the standard scope of documentation for the engine-generator set. The following documents are supplied in addition:
- Test records
- Spare parts catalog
- Wiring diagrams

### Definition of MTU

MTU refers to Rolls-Royce Power Systems AG and MTU Friedrichshafen GmbH or an affiliated company pursuant to Section §15 AktG (German Stock Corporation Act) or a controlled company (joint venture).

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# 2 Safety

## 2.1 Important provisions for all products

### General

This product may pose a risk of injury or damage in the following cases:
- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

### Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by MTU come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

### Emissions regulations and emissions label

#### Responsibility for compliance with emissions regulations

Modification or removal of any mechanical/electronic components or the installation of additional components including the execution of calibration processes that might affect the emissions characteristics of the product are prohibited by emissions regulations. Emissions-related components must only be serviced, exchanged or repaired if the components used for this purpose are approved by the manufacturer.

Noncompliance with these specifications will invalidate the design type approval or certification issued by the emissions regulation authorities. The manufacturer does not accept any liability for violations of the emissions regulations.

The product must be operated over its entire life cycle according to the conditions defined as "Intended use" (→ Page 9).

#### Emissions certification applicable to engines with EPA Nonroad Tier 4 emissions certification in accordance with 40 CFR 1039 and with EPA Locomotive emissions certification in accordance with 40 CFR 1033

Extract from the standard:

Failing to follow these instructions when installing a certified engine in a piece of nonroad equipment violates federal law (40 CFR 1068.105(b)), subject to fines or other penalties as described in the Clean Air Act.

Extract from the standard:

If you install the engine in a way that makes the engine's emission control information label hard to read during normal engine maintenance, you must place a duplicate label on the equipment, as described in 40 CFR 1068.105.

When fitting the second label, the requirements of 40 CFR 1068.105(c) must be followed and observed. This paragraph describes the process for requesting and fitting the label, the documentation obligations and storage obligations for the required documents.

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## Replacing components with emissions labels

On all engines from MTU fitted with emissions labels, these labels must remain on the engine throughout its operational life.

Exception: Engines used exclusively in land-based, military applications other than by US government agencies.

Please note the following when replacing components with emissions labels:
- The relevant emissions labels must be affixed to the spare part.
- Emissions labels shall not be transferred from the replaced part to the spare part.
- The emissions labels must be removed from the replaced part and destroyed.

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## 2.2 Gas systems – Intended use

### Intended use

The machine is intended to be used exclusively for electrical power generation and waste heat utilization in stationary operation, either in sufficiently ventilated rooms or in container solutions.

A gas/air mixture is fed to the internal combustion reciprocating piston engine. This is converted into mechanical work in the individual cylinders through combustion. The flywheel of the engine drives the synchronous generator connected via an elastic coupling, resulting in a conversion to electric energy. The engine and the generator are joined to the base frame with resilient, vibration-damped elements. For mains frequency 60 Hz, depending on the series either a gearbox is used between the engine and generator (Series 4000) or the engine speed is increased accordingly (Series 400). The waste heat generated during the combustion process is dissipated to the surroundings via coolers (power generator) and/or made available to the customer via heat exchangers (combined heat and power plant).

| Area of application | Applicable | Comment |
|---|---|---|
| Industry | Yes | – |
| Business/Service | Yes | In some individual cases, however, such as with swimming pool facilities, schools, heating stations for apartment buildings and the like, no end user in the classic sense is involved |
| Household | No | Operation is excluded |
| Potentially explosive area | No | Operation is not permitted in potentially explosive areas |
| Underground | No | Operation is excluded |
| Mobile use | No | Operation is excluded |

*Table 1: Area of application of machine*

There are 3 operating modes of the machine:
- Mains parallel operation
- Mains failure operation
- Isolated operation

### Foreseeable misuse

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper use.

### Operation

Operation is not permissible if the physical conditions specified on the technical data sheet or on/in other documents (concerning temperature, pressure, site altitude, etc.) are not complied with. Operation is not permissible without protective equipment, usage of approved spare parts, exclusion of conversions, warning of foreseeable misuse.

Operation is only permissible with the use of fluids and lubricants (natural gas, biogas, quality-compliant gas, quality-compliant water, lubricants) in accordance with the (→ Fluids and Lubricants Specifications).

Operation is only permissible with preservation agents approved by MTU in accordance with the (→ Preservation and Represervation Specifications).

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version of these are adopted. The latest version can be found at:
- http://www.mtu-solutions.com

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## Surroundings, climate, temperature

The weather plays no role with regard to the intended use of the engine-generator set/module provided that operation takes place inside an installation room. For container applications, however, the permissible outdoor temperatures at the installation site play a decisive role. The product-specific ambient conditions such as climate and temperature are indicated in the technical description.

## Prerequisite for proper use

Fulfillment of all the requirements set out in the “Installation conditions” document.

Any use that does not fall within the described limits for use is regarded as improper.

Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.

## Modifications or conversions

Unauthorized changes to the product breach the conditions concerning its intended use and compromise safety.

Only modifications or conversions expressly authorized by the manufacturer are regarded as compliant in this sense. The manufacturer shall not be held liable for any damage resulting from unauthorized modifications or conversions.

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